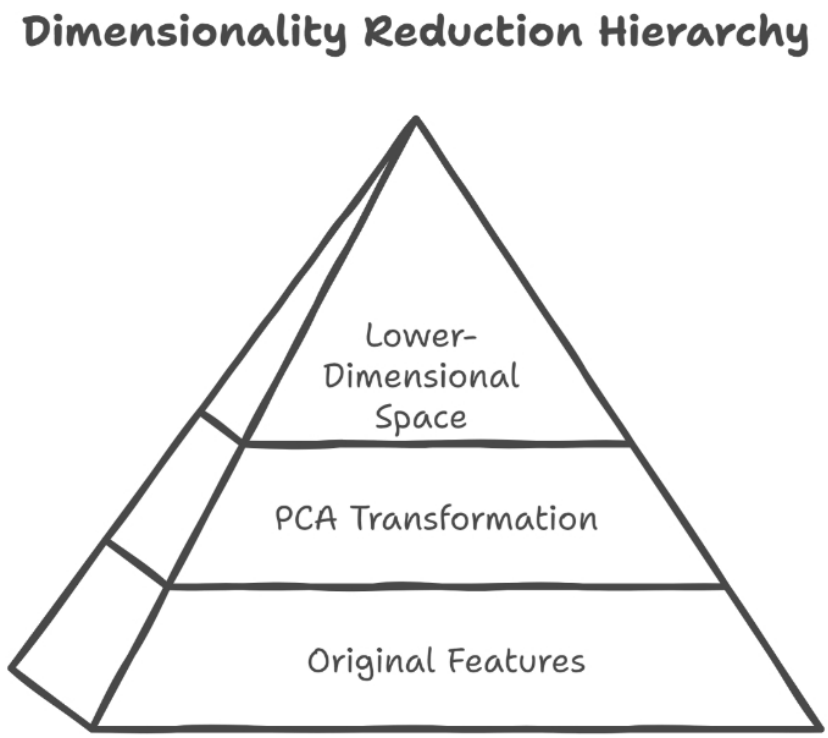
Feature Importance from Models

Tree-based models, such as Random Forest and Gradient Boosting, provide feature importance scores. These scores can be used to select significant features: from sklearn.ensemble import RandomForestClassifier

```
model = RandomForestClassifier()
model.fit(data, target)
feature_importances = model.feature_importances_
```

Dimensionality Reduction

Dimensionality reduction techniques, such as Principal Component Analysis (PCA), transform features into a lower-dimensional space while preserving variance:




```
from sklearn.decomposition import PCA

pca = PCA(n_components=2)
data_reduced = pca.fit_transform(data)
```

Feature engineering is an art and science that transforms raw data into insightful features, enabling machine learning models to excel. With Scikit-